Unit 7. Universal gates and Physics of integrated Circuits

Logic gates:-

An electronic circuit which makes logic decisions or perform logic functions and hence has particular output for unique combination of input is called logic gates. They are basic building blocks of digital systems.

The input or output of logic gates is stated by either '0' or '1' state. '0' has meaning to low voltage,false statement, off state, open circuit e.t.c. while '1' refer for high voltage, true statement, on state, close circuit e.t.c. . Logic gates may have one to many input system but only has one output. Logic gates work on the logical algebra developed by George Boole named Boolean algebra.

Boolean algebra is an algebra of logic basically used in logic systems in order to relate inputs to outputs. This algebra mainly use two variable '1' or '0' but no negative and fractional number logically.

There is mainly fundamentally three types of logic gates from which others types can be derived .
Logic gates are of follow types.
1. Fundamental logic gates.
2. Compound logic gates.

1. Fundamental logic gates.

There are three fundamental logic gates:-

by S.K Sir

a. OR logic gate:-

An OR logic gate is one in which its output is on high state if any or all of inputs are in high state. So, this gate is also named as all or any gate. It is '1' sensitive gate as its output result to '1' if any one of its input is '1'.

The boolean algebra of OR gate is $Y=A+B$; for two input system.

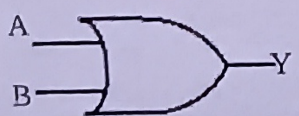


Fig: Logic symbol of OR gate.

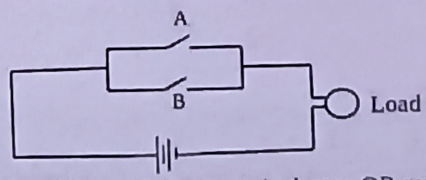


Fig: Parallel switch circuit equivalent to OR gate.

This logic gate can also be equivalent to parallel switch circuit as the load will glow if any one of switch is closed.

An OR logic circuit can be developed by connecting two diodes D_1 and D_2 in parallel with common their N-region and grounded through R-resistance as shown in fig.

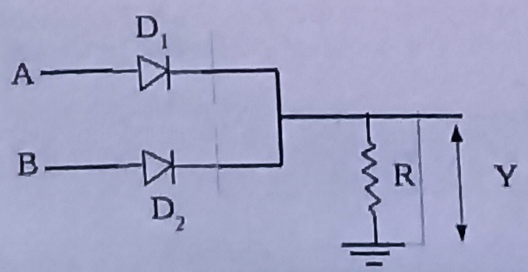


Fig: OR gate using diode.

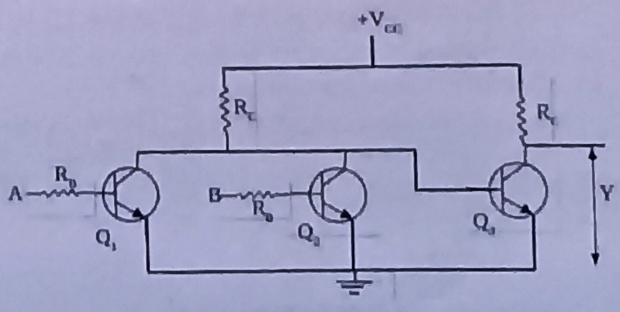


Fig: OR-gate using NPN transistor.

For A and B be two inputs of gates and Y be output across load resistance 'R' and further, 0V or +5V are two possibles inputs voltages representing logic states '0' or '1' respectively from ground (-Ve terminal) and positive terminal (+V_{cc}) of power source .

Then following results are obtained for possible inputs combination for OR logic gate.

Case I: When A=0 , B=0; i.e. both inputs are 0V then diodes D_1 and D_2 both are reversed biased and do not conduct current. Hence, the potential drop across load resistance is zero. So output is of '0' state.

Case II: When A=0 , B=1; then diodes D_1 is in reverse bias and D_2 gets forward bias and So D_2 diode conduct current while D_1 is at off state i.e. do not conduct current. Hence, the certain potential drop occurs across load resistance as output. So output is of '1' state.

Case III: When A=1 , B=0; then diodes D_1 is in forward bias and D_2 gets reverse bias and So D_1 diode conduct current while D_2 is at off state i.e. do not conduct current. Hence, the certain potential drop occurs across load resistance as output. So output is of '1' state.

Case IV: When A=1 , B=1; i.e. both inputs are +5V then diodes D_1 and D_2 both are forward biased and both conduct current. Hence, the certain potential drop occurs across load resistance as output. So output is of '1' state.

Truth table:- A table showing all possible inputs combination and their corresponding outputs in a a tabular form is known as truth table of given logic gates.

Inputs		Output Y=A+B
A	B	
0	0	0
0	1	1
1	0	1
1	1	1

Fig: Truth table of OR logic gate

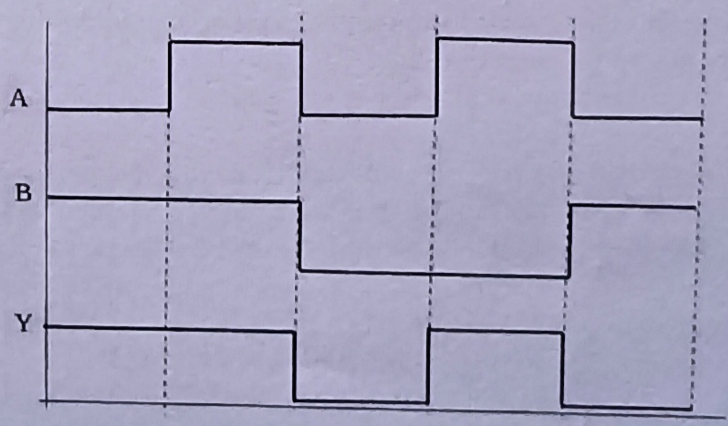


Fig: Output wave form of OR-gate.

b. AND logic gate:-

An AND logic gate is one in which its output is on high state if and only if all of inputs are in high state. So, this gate is also named as all or none gate. It is '0' sensitive gate as its output result to '0' if any one of its input is '0'.

The boolean algebra of AND gate is Y=A.B ; for two input system.

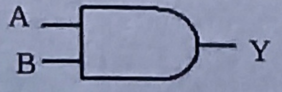


Fig: logic symbol of AND gate.

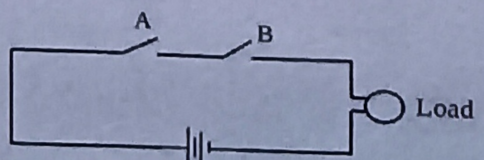
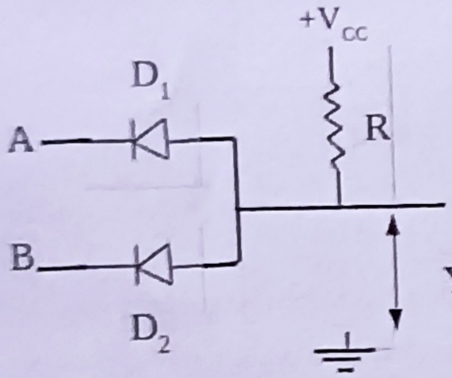


Fig: Series switch circuit equivalent to AND gate.

This logic gate can also be equivalent to series switch circuit as the load will glow if and only if both switch is closed.

An AND logic circuit can be developed by connecting two diodes D_1 and D_2 in parallel with common their P-region and is supplied with high $+V_{cc}$ potential through 'R' resistance as shown in fig.



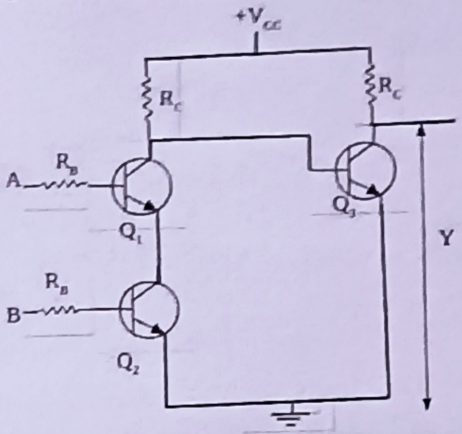


Fig: AND-gate using NPN transistor.

Fig: AND gate using diodes.

For A and B be two inputs of gates and Y be output across the common point 'p' of diodes to ground. Further, 0V or +5V are two possibles inputs voltages representing logic states '0' or '1' respectively from ground (-Ve terminal) and positive terminal (V_{cc}) of power supply .

Then following results are obtained for possible inputs combination for AND logic gate.

Case I: When A=0 , B=0; i.e. both inputs are 0V then diodes D_1 and D_2 both are forward biased and conduct current. Hence, the potential drop across point 'p' to ground 'G is zero. So output is of '0' state.

Case II: When A=0 , B=1; then diodes D_1 is in forward bias and D_2 gets reversed bias and So D_1 diode conducts current while D_2 is at off state i.e. do not conduct current. Hence, the potential difference between point 'P' and ground is zero. So output is again at low '0' state.

Case III: When A=1 , B=0; then diodes D_2 is in forward bias and D1 gets reverse bias and So D_2 diode conduct current while D_1 is at off state i.e. do not conduct current. Hence, the potential difference between point 'P' and ground is zero. So output is again at low '0' state.

Case IV: When A=1 , B=1; i.e. both inputs are +5V then diodes D_1 and D_2 both are reversed biased and both donot conduct current. Hence, there is potential difference between point 'P' and ground .So output is of high '1' state.

Truth table:- A table showing all possible inputs combination and their corresponding outputs in a a tabulated form is known as truth table of given logic gates.

Inputs		Output Y=A.B
A	B	
0	0	0
0	1	0
1	0	0
1	1	1

Fig: Truth table of OR logic gate

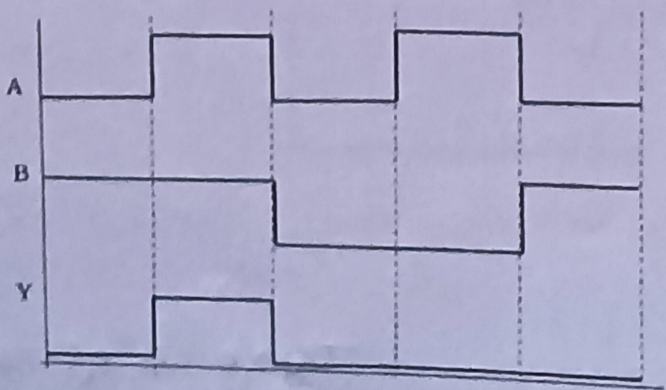


Fig: Output wave form of AND-gate.

c. NOT logic gate:-

A NOT logic gate is one in which its output is always inverse to its input logic state hence it is also named as invert gate. NOT gate is only one gate that has one input system with one output.

The boolean algebra of NOT gate is $Y=\overline{A}$

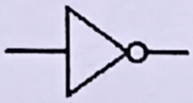


Fig: Logic symbol of AND gate.

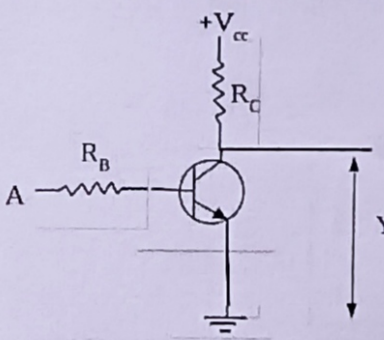


Fig: NOT gate using NPN transistor.

As logic symbol of NOT gate contains bubble part on its output side representing complement of input at output So, it is also named as bubbled gate.

For, 0V or +5V are two possibles inputs states of 'A' representing logic states '0' or '1' respectively . The ground refers for 0V potential while V_{cc} for higher +5V potential.

Following results are obtained for possible inputs for NOT logic gate.

Case I: When A=0 i.e. 0 V is input to a base of transistor ,base-emitter junction of transistor is reverse bias and hence is in open state. So current from V_{cc} can't grounded through transistor. Thus, the potential difference appears across collector to ground at the output representing a high state '1'.
Hence, for low input, it has high output.

Case II: When A=1 i.e. +5V is input to a base of transistor ,base-emitter junction of transistor is forward bias and hence is in close state. So current from V_{cc} flows through transistor to ground. Thus, there is no more potential difference appears across collector to ground at the output. This represents a low state '0'.
Hence, for high input, it has low output.

Truth table:- A table showing all possible inputs combination and their corresponding outputs in a a tabulated form is known as truth table of given logic gates.

Input A	Output $Y=\overline{A}$
0	1
1	0

Fig: Truth table of NOT logic gate

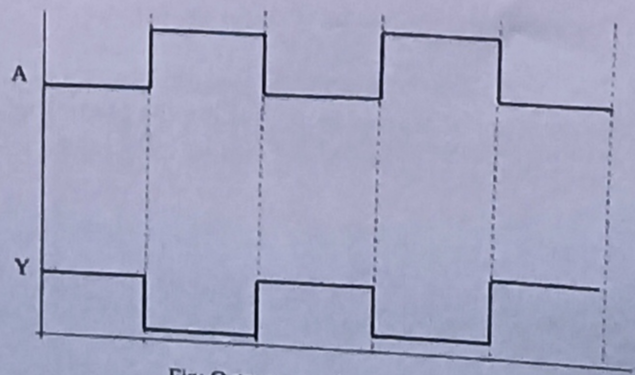


Fig: Output wave form of NOT-gate.

b. Compound (Combination of Fundamental gates) Logic gates:-

Compound logic gates are gates formed by the combination of fundamental gates. The important compound logic gates are NOR, NAND, Ex-OR, Ex-NOR gates e.t.c.

1. NOR gate:-

An NOR logic gate is one in which its output is on high state if and only if all of inputs are in low state. So, it is '1' sensitive gate as its output result to '0' if any one of its input is '1'. This gate is complement of output of OR gate i.e. output of OR gate is followed by NOT gate.

The boolean algebra of NOR gate is $Y=\overline{A+B}$; for two input system.

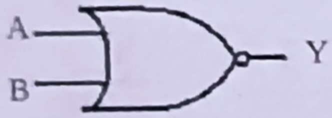


Fig: Logic symbol of NOR gate.

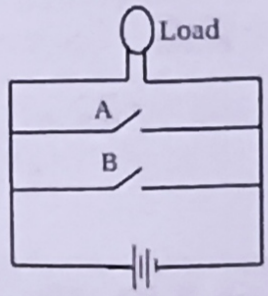


Fig: Parallel switch circuit across cell equivalent to NOR gate.

The output of NOR gate resembles with output of AND gate and is equivalent to parallel switch circuit across a source and load as shown in fig.

The circuit diagram using diodes shows that the output of OR gate is inverted using NOT gate transistor circuit.

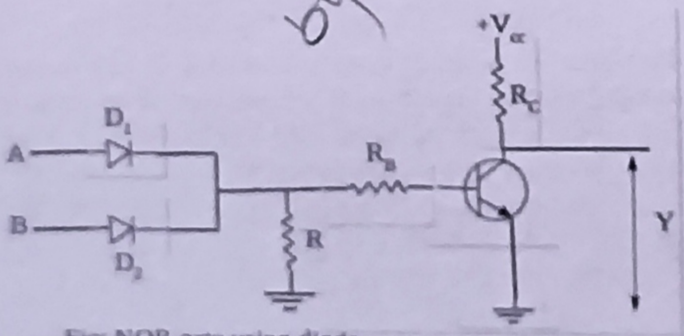


Fig: NOR gate using diode.

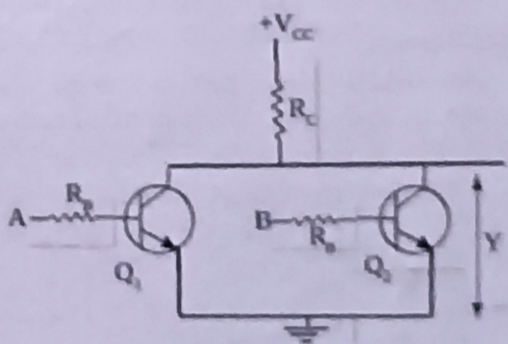


Fig: NOR-gate using NPN transistor.

For A and B be two inputs of gates and Y be output across the collector to ground of transistor and further, 0V or +5V are two possibles inputs voltages representing logic states '0' or '1' respectively from ground (-Ve terminal) and positive terminal (V_{cc}) of power supply .

Then following results are obtained for possible inputs combination for NOR logic gate.

Case I: When A=0 , B=0; i.e. both inputs are 0V then diodes D_1 and D_2 both are reversed biased and do not conduct current. So that output of OR gate is low or to say input of NOT gate is low and further which is reversed by NOT gate to high state as its output. So output is of '1' state.

Case II: When A=0 , B=1; then diodes D_1 is in reverse bias and D_2 gets forward bias and So D_2 diode conduct current while D_1 is at off state i.e. do not conduct current. So that output of OR gate is high or to say input of NOT gate is high and further which is reversed by NOT gate to low state as its output. So output is of '0' state.

Case III: When A=1 , B=0; then diodes D_1 is in forward bias and D_2 gets reverse bias and So D_1 diode conduct current while D_2 is at off state i.e. do not conduct current. So that output of OR gate is high or to say input of NOT gate is high and further which is reversed by NOT gate to low state as its output. So output is of '0' state.

Case IV: When A=1 , B=1; i.e. both inputs are +5V then diodes D_1 and D_2 both are forward biased and both conduct current. So that output of OR gate is high or to say input of NOT gate is high and further which is reversed by NOT gate to low state as its output. So output is of '0' state.

Truth table:- A table showing all possible inputs combination and their corresponding outputs in a a tabulated form is known as truth table of given logic gates.

Inputs		Output $Y=\overline{A+B}$
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

Fig: Truth table of NOR logic gate

by S.G. Sir

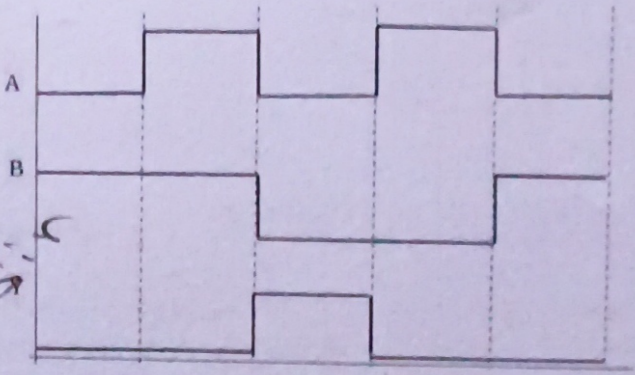


Fig: Output wave form of NOR-gate.

2. NAND gate:-

An NAND logic gate is one in which its output is on high state if and only if any one of inputs are in low state. So, it is '0' sensitive gate as its output result to '1' if any one of its input is '0'. This gate is complement of output of AND gate i.e. output of AND gate is followed by NOT gate.The boolean algebra of NAND gate is $Y=\overline{A.B}$; for two input system.

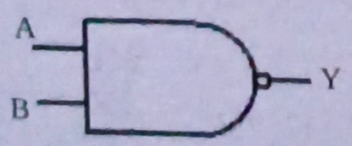


Fig: Logic symbol of NAND gate.

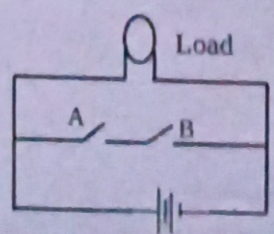


Fig: Series switch circuit across cell equivalent to NAND gate.

The output of NAND gate resembles with output of OR gate and is equivalent to series switches circuit across a source and load as shown in fig.
The circuit diagram using diodes shows that the output of AND gate is inverted using NOT gate transistor circuit.

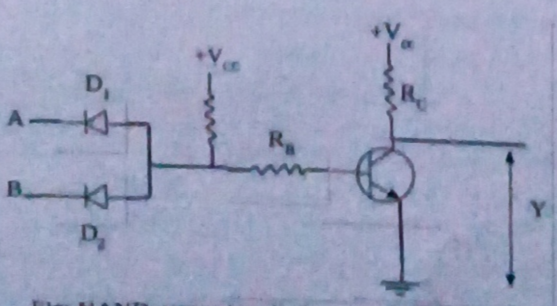


Fig: NAND gate using diode.

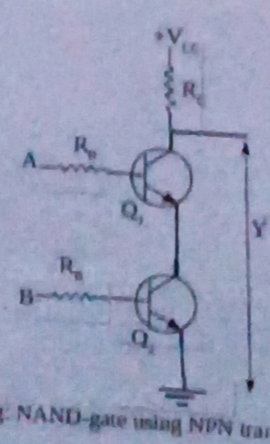


Fig: NAND-gate using NPN transistor.

For A and B be two inputs of gates and Y be output across the collector to ground of transistor and further, 0V or +5V are two possibles inputs voltages representing logic states '0' or '1' respectively from ground (-Ve terminal) and positive terminal (V_{cc}) of power supply .

Then following results are obtained for possible inputs combination for NAND logic gate.

Case I: When A=0 , B=0; i.e. both inputs are 0V then diodes D_1 and D_2 both are forward biased and conduct current. So that output of AND gate is low or to say input of NOT gate is low and further which is reversed by NOT gate to high state as its output. So output is of '1' state.

Case II: When A=0 , B=1; then diodes D_1 is in forward bias and D_2 gets reverse bias and So D_1 diode conduct current while D_2 is at off state i.e. do not conduct current. So that output of AND gate is low or to say input of NOT gate is low and further which is reversed by NOT gate to high state as its output. So output is of '1' state.

Case III: When A=1 , B=0; then diodes D_1 is in reverse bias and D_2 gets forward bias and So D_1 is at off state i.e. do not conduct current while diode D_2 do conduct current. So that output of AND gate is low or to say input of NOT gate is low and further which is reversed by NOT gate to high state as its output. So output is of '1' state.

Case IV: When A=1 , B=1; i.e. both inputs are +5V then diodes D_1 and D_2 both are reverse biased and both don't conduct current. So that output of AND gate is high or to say input of NOT gate is high and further which is reversed by NOT gate to low state as its output. So output is of '0' state.

Truth table:- A table showing all possible inputs combination and their corresponding outputs in a a tabulated form is known as truth table of given logic gates.

Inputs		Output
A	B	$Y=\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

Fig: Truth table of NAND logic gate

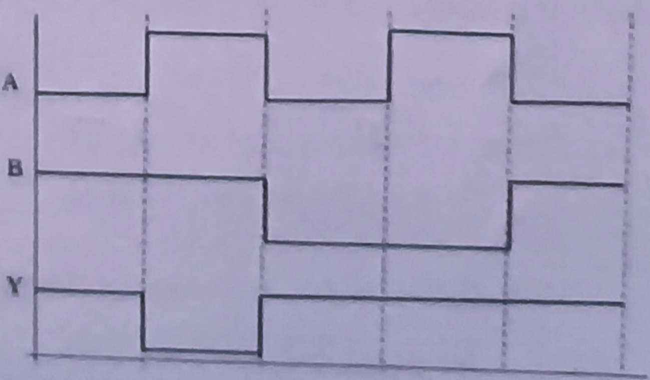


Fig: Output wave form of NAND-gate.

 # De- Movier's Theorem:-

1. First theorem :- The complement of a sum is equals to the product of complements.
   i.e. $\overline{A+B}=\overline{A}.\overline{B}$
   NOR gate is equal to bubbled inputs of AND gate.

2. Second theorem :- The complement of a product is equals to the sum of complements.
   i.e. $\overline{A.B}=\overline{A}+\overline{B}$
   NAND gate is equal to bubbled inputs of OR gate.

Universal Gates:-

Those logic gates with which all the fundamental logic gates OR, AND, and NOT can be derived are called universal gates. Logic gates NAND and NOR can be used to make fundamental logic gates so these are called universal gates.
The conversion of universal gates.

a. NAND gate as universal gate:-

A NAND gate can be use in order to perform all logic operation of OR gate, AND gate and NOT gate which are explained as below;

i. NAND gate as a NOT gate:-

A NOT gate can be obtained from a NAND gate by simply connecting its two inputs together as shown in fig

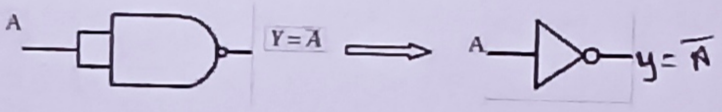


Fig: NOT gate using NAND gate.

The truth table of NOT gate using NAND gate is

Inputs A=B	Output $Y=\overline{A}$
0	1
1	0

by S.G Sir

ii. NAND gate as a AND gate:-

AND gate can be formed by using two NAND gates, one as NAND gate followed by other as NOT configuration in series connection as shown in figure.

$\overline{\overline{A \cdot B}} = A \cdot B$

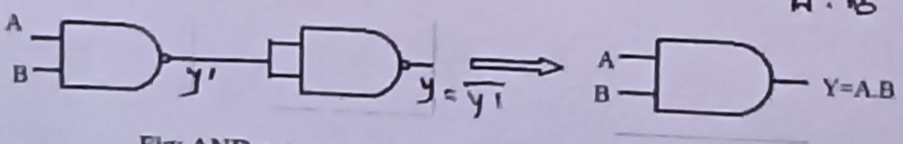


Fig: AND gate using NAND gate.

The truth table of AND gate obtained using NAND gate is

Inputs			Output
A	B	$Y_1=\overline{A.B}$	$Y=\overline{Y_1}$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

iii. NAND gate as a OR gate:-

Three NAND gates are used to perform logic operation as of OR gate. Two are use as NOT gate and are connected in inputs of 3rd NAND gate in order to invert the inputs it. The output of 3rd NAND gate is equivalent to that of OR gate. $\overline{\overline{A}\cdot\overline{B}} = A+B$

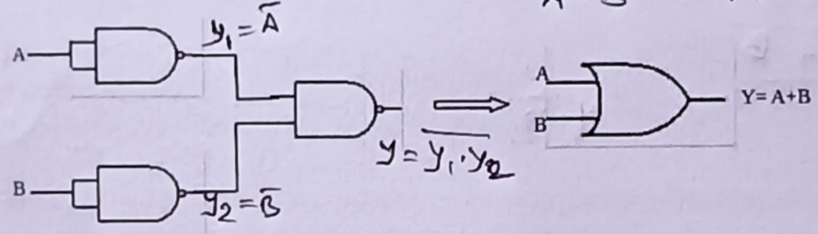


Fig: OR gate using NAND gate.

The truth table of OR gate obtained using NAND gate is

Inputs		$Y_1=\overline{A}$	$Y_2=\overline{B}$	Output $Y=\overline{Y_1 . Y_2}$
A	B			
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

b. NOR gate as universal gate:-

A NOR gate can ~~be use~~ in order to perform all logic operation of OR gate, AND gate and NOT gate which are explained as below;

i. NOR gate as a NOT gate:-

A NOT gate can be obtained from a NOR gate by simply connecting its two inputs together as shown in fig

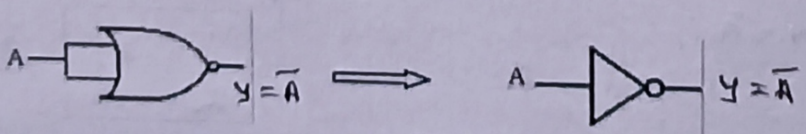


Fig: NOT gate using NOR gate.

The truth table of NOT gate using NAND gate is

Inputs A=B	Output $Y=\overline{A}$
0	1
1	0

ii. NOR gate as a OR gate:-

OR gate can be obtained by using two NOR gates, one as NOR gate followed by other as NOT configuration in series connection as shown in figure.

$$A + B = \overline{\overline{A+B}}$$

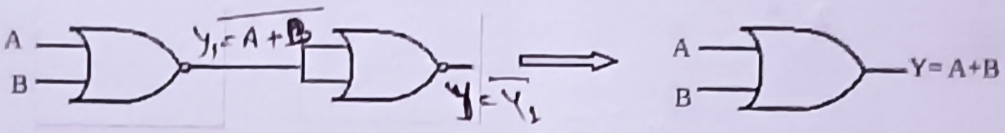


Fig: OR gate using NOR gate.

The truth table of OR gate using NOR gate is

Inputs		$Y_1=\overline{A+B}$	Output $Y=\overline{Y_1}$
A	B		
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

iii. NOR gate as a AND gate:-

Three NOR gates are used to perform equivalent logic operation as of AND gate. Two are use as NOT gate and are connected in inputs of 3rd NOR gate in order to invert the inputs . The output of 3rd NOR gate is equivalent to that of AND gate.

$$\overline{\bar{A} + \bar{B}} = A \cdot B$$

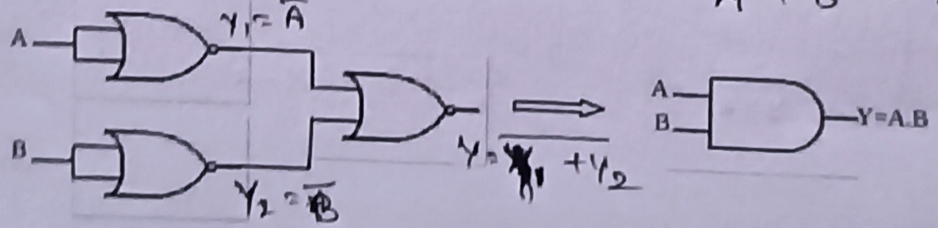


Fig: AND gate using NOR gate.

The truth table of AND gate using NOR gate is

Inputs		$Y_1=\overline{A}$	$Y_2=\overline{B}$	Output $Y=\overline{Y_1+Y_2}$
A	B			
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1